V5A14 Representation of p-adic Groups

Lectures Given By
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1 Preliminaries

Let F be a non-Archimedian local field (i.e., $F/\mathbb{Q}_p$ for some finite extension of a p-adic numbers or $F = \mathbb{F}_q((t))$ where q is a power of p). We denote $\mathcal{O}_F \subseteq F$ for the ring of integers.

Example 1.1. *Examples of pairs $(F, \mathcal{O}_F)$ as above are $(\mathbb{Q}_p, \mathbb{Z}_p)$ and $(\mathbb{F}_q((t)), \mathbb{F}_q[\![t]\!])$.*

We will denote ϖ for the uniformizer (i.e., a generator of the unique maximal ideal $\mathfrak{p} \subseteq \mathcal{O}$).

Example 1.2. *In the first pair of Example 1.1, we have $\varpi = p$, hence*

$$\mathcal{O}_F/\varpi\mathcal{O}_F = \mathbb{Z}_p/(p)\mathbb{Z}_p \cong \mathbb{F}_p.$$

In the second pair, $\varpi = t$, hence

$$\mathcal{O}_F/\varpi\mathcal{O}_F = \mathbb{F}_q[\![t]\!]/(t)\mathbb{F}_q[\![t]\!] \cong \mathbb{F}_q.$$

The topology on $\mathrm{GL}_n(F)$ is defined by the neighborhood basis around the origin. Explicitly, this is constructed by

$$I_n + \varpi \operatorname{Mat}_{n\times n}(\mathcal{O}) \supseteq I_n + \varpi^2 \operatorname{Mat}_{n\times n}(\mathcal{O}) \supseteq \cdots$$

Definition 1.1. *Let $G \subseteq \mathrm{GL}_n(F)$ be a group. A smooth representation of G is a pair (π, V) such that*

1. *V is a complex vector space.*
2. *$\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ is a group homomorphism.*
3. *$\forall v \in V$, $\mathrm{Stab}_G(v) := \{g \in G \mid \pi(g)(v) = v\}$ is open.*

Example 1.3. *For $\mathrm{GL}_1(\mathbb{Q}_p) = \mathbb{Q}_p^\times$, $V = \mathbb{C}$ the we can take,*

$$\pi(g) = |g|_p = p^{-\mathrm{val}(g)} \in \mathrm{Aut}(\mathbb{C}) \cong \mathbb{C}^\times.$$

Definition 1.2. *A representation (π, V) is irreducible if (π, V) has precisely two sub-representations (i.e., the trivial representations $\{0\}$ and itself).*

Remark 1.1. *By the definition above, $\{0\}$ is not an irreducible representation.*

Remark 1.2. *All irreducible smooth representations of $\mathbb{Q}_p^\times$ are one dimensional. This follows from the fact that all irreducible representations admits a character which agree with the original representation on the center. In our case, as $\mathbb{Q}_p^\times$ is commutative hence must be the whole.*

1.1 Linear Algebraic Groups

Definition 1.3. *Let k be an algebraically closed field. A linear algebraic group over k is a reduced Zariski-closed subgroup of GL_n over k for some $n \in \mathbb{N}$.*

Equivalently, linear algebraic groups over k are smooth affine group schemes of finite type over k.

Notation 1.1. *From now on, $\mathbb{G}$ will denote the linear algebraic groups and $G = \mathbb{G}(k)$ the k-points of $\mathbb{G}$.*

Example 1.4. $\mathrm{GL}_n(k), \mathrm{SL}_n(k), \mathrm{O}_n(k), \mathrm{Sp}_{2n}(k)$ *are linear algebraic groups and, $B(k) \subseteq \mathrm{GL}_n(k)$, upper triangular matrices*

Definition 1.4. *$G \subseteq \mathrm{GL}_n(k)$ is unipotent if G is conjugate to a subgroup of upper triangular matrices with 1-s in the diagonal.*

Definition 1.5. *The unipotent radical (denoted by $(RG)_u$ or R_uG) of a linear algebraic group is the maximal connected unipotent normal subgroup of G.*

Definition 1.6. *An algebraic group G is reductive if its unipotent radical is trivial.*

Example 1.5. $\mathrm{GL}_n(k), \mathrm{SL}_n(k), \mathrm{SO}_n(k), \mathrm{Sp}_{2n}(k)$ *are all reductive. $U_n(k)$ and $B(k)$ are not.*

Definition 1.7. *A linear algebraic group G is called a torus if $G \cong (k^\times)^n$ for some $n \in \mathbb{N}$.*

Theorem 1.1. *All maximal tori in an linear algebraic group G are conjugate.*

Notation 1.2. *T will be a maximal torus in G. Then we will denote $\mathfrak{g} = \mathrm{Lie}(G) := T_1(G)$ with Lie algebra structure.*

Example 1.6.

Linear Algebraic Groups	*Lie Algebra*	*Commutator*
GL_n	$\mathrm{Mat}_{n\times n}$	$[A, B] = AB - BA$
SL_n	$\{A \in \mathrm{Mat}_{n\times n} \mid \mathrm{tr}(A) = 0\}$	$[A, B] = AB - BA$
Sp_{2n}	$\left\{ A \in \mathrm{Mat}_{n\times n} \;\middle|\; \mathrm{tr}(A) = 0, JA + {}^tAJ = 0 \right\}$	$[A, B] = AB - BA$

Remark 1.3. *G acts on $\mathfrak{g}$ via the adjoint action. Precisely, this is the derivation of the conjugation action on itself. Explicitly, for $g \in \mathrm{GL}_n(k)$ and $A \in \mathrm{Mat}_{n\times n}(k)$, we define,*

$$\mathrm{Ad}(g)(A) = gAg^{-1}.$$

Hence T also acts on $\mathfrak{g}$ and such induced actions are diagonalizable.

Example 1.7.

$$\mathrm{diag}(t_1, \cdots, t_n)E_{ij} = t_i t_j^{-1} E_{ij}.$$

Definition 1.8. *$X^*(T) = \mathrm{Hom}_k(T, \mathbb{G}_m)$ is called the characters of T.*

Example 1.8. *$X^*(\mathbb{G}_m(k)) \cong \mathbb{Z}$. Explicitly, this isomorphism is $k^\times \ni x \mapsto x^n \in k^\times$ for some n.*

From now on, G will denote the connected reductive group.

Proposition 1.1. *For $\mathfrak{g} = \mathrm{Lie}(G)$. We have,*

$$\mathfrak{g} = \bigoplus_{\alpha \in X^*(T)} \mathfrak{g}_\alpha,$$

where $\mathfrak{g}_\alpha$ is the subspace such that T acts via the character α. Explicitly, this is

$$\mathfrak{g}_\alpha := \{X \in \mathfrak{g} \mid \forall t \in T, \mathrm{Ad}(t)(X) = \alpha(t)X\}.$$

The root of G is

$$\Phi(G) = \Phi(G;T) := \{\alpha \in X^*(T)\backslash\{0\} \mid \dim \mathfrak{g}_\alpha \neq 0\}.$$

Remark 1.4. $\dim \mathfrak{g}_\alpha = 1$ *for all $\alpha \in \Phi(G)$ and $\Phi(G) \subseteq X^*(T) \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$ which is a root system. Such root systems are classified via Dynkin diagrams.*

The root system $\Phi(G) = \Phi(G;T)$ is independent of T up to isomorphisms and $G/Z(G)$ is uniquely determined by $\Phi(G)$.

Example 1.9. *For $G = \mathrm{GL}_n(k)$ and $\mathfrak{g} = \mathfrak{gl}_n$, we have,*

$$\mathrm{Mat}_{n\times n}(k) = \bigoplus_{\alpha \in X^*(T)} \mathfrak{g}_\alpha.$$

This is

$$T \oplus \bigoplus_{i \neq j} \langle E_{ij} \rangle.$$

Hence,

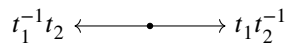
$$\Phi(\mathrm{GL}_n) = \{\mathrm{diag}(t_1, \cdots, t_n) \mapsto t_i t_j^{-1} \mid 1 \leq i, j \leq n, i \neq j\}.$$

For $n = 2$,

$$X^*(T) = \mathbb{Z}t_1 \oplus \mathbb{Z}t_2.$$

The root system is

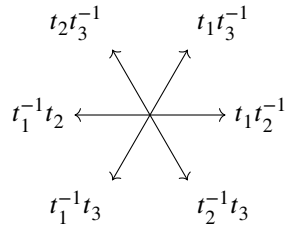
$$\Phi(\mathrm{GL}_2) = \{\mathrm{diag}(t_1, t_2) \mapsto t_1 t_2^{-1}, \mathrm{diag}(t_1, t_2) \mapsto t_2 t_1^{-1}\}.$$



For $n = 3$,

$$X^*(T) = \mathbb{Z}t_1 \oplus \mathbb{Z}t_2 \oplus \mathbb{Z}t_3.$$

In a graph, this is



Remark 1.5. *A reductive group is called semisimple if its center is finite. Root system does not determine the groups, however, they do determine groups modulo their center.*

Definition 1.9. *For a root system Φ, a subset $\Delta \subseteq \Phi$ is called a basis of it if every element of the root system can be written as a linear combination of elements of Δ*

$$\sum_{\alpha\in\Delta} a_\alpha \alpha,$$

such that either a_α are all non-negative or they are all non-positive.

Example 1.10. *For $G = \mathrm{GL}_n$, we have,*

$$\Delta(\mathrm{GL}_n) = \left\{ \begin{pmatrix} t_1 & & \\ & \ddots & \\ & & t_n \end{pmatrix} \mapsto t_i t_{i+1}^{-1} \;\middle|\; 1 \le i < n \right\} = \Delta(\mathrm{SL}_n).$$

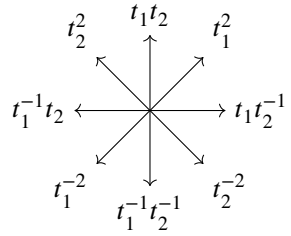
Example 1.11. *For $G = \mathrm{Sp}_{2n}$ and a torus*

$$T = \{\mathrm{diag}(t_1, \cdots, t_n, t_1^{-1}, \cdots, t_n^{-1}) \mid t_1, \cdots, t_n \in F^\times\},$$

the root system is given by

$$\Phi(\mathrm{Sp}_{2n})\{\mathrm{diag}(t_1, \cdots, t_n, t_1^{-1}, \cdots, t_n^{-1}) \mapsto t_i^{\pm 1} t_j^{\pm 1}, t_i^{\pm 2} \mid 1 \le i \ne j \le n\}.$$

In particular, for $n = 2<$, we have the following diagram.



The basis is given by

$$\Delta(\mathrm{Sp}_{2n}) = \{\mathrm{diag}(t_1, \cdots, t_n, t_1^{-1}, \cdots, t_n^{-1}) \mapsto t_i^1 t_{i+1}^{-1}, t_i^2 \mid 1 \leq i < n\}.$$

Definition 1.10. *A subgroup P of a reductive group G is called parabolic if it is Zariski-closed and and G/P is projective.*

Definition 1.11. *A Borel subgroup of a reductive group G is a maximal connected solvable Zariski closed subgroup of G.*

Remark 1.6. *A subgroup is parabolic if and only if it is Zariski-closed and contains some Borel subgroup.*

Theorem 1.2. *All Borel subgroups are conjugate to one another.*

Example 1.12. *In GL_n, a subgroup of upper-triangular matrices is Borel. Hence the subset all block-upper-triangular matrices is parabolic.*

Theorem 1.3. *There is a one-to-one correspondence between following sets.*

$$\left\{ \begin{matrix} \textit{Parabolic subgroups containing} \\ \textit{a fixed Borel subgroup } B \end{matrix} \right\} \overset{1:1}{\leftrightarrow} \{\textit{Subsets of a basis } \Delta \textit{ of the root system } \Phi(\mathrm{GL}_n)\}.$$

Let $T \subseteq B$ be a torus contained in a Borel subgroup. Then we have

$$\mathrm{Lie}(B) = \underbrace{\mathfrak{Z}}_{:=\mathrm{Lie}(T)} \bigoplus_{\alpha \in \Phi^+(G)} \mathfrak{g}_\alpha,$$

where

$$\Phi^+(G) := \left\{ \sum_{\alpha_i \in \Delta} n_i \alpha_i \;\middle|\; \forall i, n_i \geq 0 \right\} \subseteq \Phi(G).$$

Example 1.13. *For $G = \mathrm{GL}_n$,*

$$\Phi^+(\mathrm{GL}_n) = \{\mathrm{diag}(t_1, \cdots, t_n) \mapsto t_i t_j^{-1} \mid 1 \leq i < j \leq n\}.$$

Observe that $t_i t_{i+1}^{-1}$ corresponds to a entry one above the diagonal. Since standard parabolic subgroups are block-upper-triangular, we can encode which blocks to include by which one-above diagonal elements to include.

That is to say under the bijection above, we have a following correspondence between parabolic subgroups and subsets of the basis.

$$P \leftrightarrow \Theta \subseteq \Delta.$$

Where $\rightarrow$ is given by

$$P \mapsto \mathrm{Lie}(P) = \mathfrak{Z} \oplus \bigoplus_{\alpha \in \Phi^+(G)} \mathfrak{g}_\alpha \oplus \bigoplus_{\alpha \in \Theta} \mathfrak{g}_{-\alpha}.$$

Clearly B corresponds to the empty subset and the whole group G corresponds to the whole basis.

Theorem 1.4. *Let G be a reductive group and P be a parabolic subgroup of it.*

1. *P is reductive $\iff$ $P = G$.*
2. *P is isomorphic to the semidirect product $M \ltimes N$ where M is a reductive subgroup and N is the unipotent radical of P. Such M is said to be a Levi subgroup. This decomposition is not unique.*

Example 1.14. *For a block-upper-triangular matices, the Levi subgroup will be the block-diagonal matrices and the unipotent radical is the subgroup whose matrices have the block diagonal all I_k for some k.*

Definition 1.12. *Let $\alpha \in \Phi(G)$, define U_α to be the unique connected T-stable Zariski-closed subgroup of G whose Lie algebra is $\mathfrak{g}_\alpha$. Such U_α is called the root subgroup.*

Example 1.15. *For $G = \mathrm{GL}_n$ and $\alpha(\mathrm{diag}(t_1, \cdots, t_n)) = t_i t_j^{-1}$, the root subgroup is the subgroup of all matrices whose diagonal entries are all 1 and only non-zero off-diagonal entry is (i, j)-th entry which hold any value. In particular, $U_\alpha \cong (k, +)$ for any root α.*

Proposition 1.2. *Let B be a Borel subgroup then we can write $B = T \ltimes U$ where U is the unipotent radical of B. Then U as a variety $\overset{\simeq}{\prod}_{\alpha \in \Phi^+(G)} U_\alpha$.*

Definition 1.13. *Let $N(T)$ be the normalizer of T, then the Weyl group W of G is $W(T) = N(T)/T$ which admits a natural action on T by conjugation.*

Example 1.16.

$$W(\mathrm{GL}_n) \cong S_n$$

where each element of S_n corresponding to the permuting the diagona entries. It can be shown that all elements of $N(T)$ are a product an element of T and a permutation matrix.

Example 1.17.

$$W(\mathrm{Sp}_{2n}) \cong S_n \rtimes (\mathbb{Z}/2\mathbb{Z})^n$$

where S_n corresponds to the permutation of the diagonal entries and $(\mathbb{Z}/2\mathbb{Z})^n$ corresponds to the swapping of the corresponding elements (i.e., one and inverse of it).

Definition 1.14. *A reflection of an element of a root system $\alpha \in \Phi$ is a linear map such that*

$$s_\alpha(\alpha) = -\alpha,$$

and all elements orthogonal to α are unchanged under s_α.

Proposition 1.3. *W is the Weyl group of $\Phi(G)$. That is W is isomorphic to the group generated by the reflections of each element $\alpha \in \Phi$.*

Theorem 1.5 (Bruhat Decomposition)**.** *For a reductive group G and its Weyl group W of a torus T, for each $w \in W$, there is n_w whose quotient by T is w. We have the following decomposition $G = \coprod_{w \in W} B n_w B$.*

Theorem 1.6.

$$Bn_wB \underbrace{\simeq}_{\text{as varieties}} B \times \{n_w\} \times \prod_{\substack{\alpha\in\Phi^+(G)\\ w^{-1}(\alpha)\notin\Phi^+(G)}} U_\alpha,$$

where the isomorphism $\leftarrow$ is given by multiplication.

2 Linear Algebraic Groups over F

As was in the previous section, we set F to be a non-archimedian local field or a finite field. (Some statements still hold without these assumptions).

Definition 2.1. *A linear algebraic group over F is a geometrically reduced closed F-subgropu of GL_n/F for some integer n. (F-subgroup is the one corresponds to a vanishing of some ideal which is defined over $\overline{F}$).*

Equivalently, a linear algebraic gropu is a smooth affine group scheme of finite type over the field F.

Definition 2.2. *A liner algebraic group $\mathbb{G}$ over F is said to be reductive if its base-change $\mathbb{G}_{\overline{F}} = \mathbb{G} \underset{F}{\times} \overline{F} := \mathbb{G} \underset{\mathrm{Spec}\, F}{\times} \mathrm{Spec}\, \overline{F}$ to the algebraic closure of F is reductive.*

Equivalently $(R\mathbb{G}_{\overline{F}})_u$ is trivial.

Example 2.1. *$\mathrm{GL}_n/F, \mathrm{SL}_n/F, \mathrm{SO}_n/F, \mathrm{Sp}_{2n}$ are all reductive.*

Definition 2.3. *An F-torus or just a torus over F is a linear algebraic group T over F such that $T_{\overline{F}}$ is a torus (i.e., $T_{\overline{F}} \cong \mathbb{G}_m^n$)*

Example 2.2. *For F'/F a finite extension, we have,*

$$\mathbb{T}(F') = \{\mathrm{diag}(t_1, t_2) \in \mathrm{GL}_2(F')\} \cong (F'^{\times})^2.$$

Example 2.3. *Let $F = \mathbb{Q}_p$ and $E = \mathbb{Q}_p(\sqrt{p})$ which is a quadratic extension,*

$$\mathbb{T}_2(F') = \left\{ \begin{pmatrix} a & b \\ pb & a \end{pmatrix} \in \mathrm{GL}_2(F') \right\}.$$

Remark 2.1. *$\mathbb{T}_2(F) \simeq E^\times$ given by*

$$\begin{pmatrix} a & b \\ pb & a \end{pmatrix} \mapsto a + b\sqrt{p}.$$

Exercise 2.1. *In the second example, $\mathbb{T}_2(\overline{F}) \simeq (\overline{F}^\times)^2$.*

Remark 2.2. *The same definition for the tosu also works for $F = \mathbb{R}$ and consider, $F' = \mathbb{R}$ or $\mathbb{C}$,*

$$\mathbb{T}_3 = \left\{ \begin{pmatrix} x & y \\ -y^x & \end{pmatrix} \right\} \subseteq \mathrm{SL}_2(F')$$

Then $\mathbb{T}_2(\mathbb{C}) \cong \mathbb{C}^\times$ *and*

$$\mathbb{T}(\mathbb{R}) = \left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \;\middle|\; \theta \in \mathbb{R} \right\}.$$

In particular $\mathbb{T}_3(\mathbb{R}) \cong S^\times \cong \mathbb{C}^\times$. *Furthermore* $(\mathbb{T}_3 \times \mathbb{T}_3)(\mathbb{R}) \cong S^1 \times S^1$ *which is a torus.*

Definition 2.4. $\mathbb{G}_m = \mathrm{GL}_1$ *(i.e.,* $\mathbb{G}_m(F') \cong (F')^\times$*) for any field extension* F'/F. *An* F*-torus* $\mathbb{T}$ *is split or* F*-split if* $\mathbb{T} \cong \mathbb{G}_m^N$ *for some* N.

Example 2.4. $\mathbb{T}_1$ *is split but* $\mathbb{T}_2$ *is not. The former is obvious and latter is due to that* $E^\times \ncong (F^\times)^n$ *for any n. This is also seen that base changin* $\mathbb{G}_m^N$ *to the algebraic closure gives* $(\overline{F})^N$.

Theorem 2.1 (Grothendieck)**.** *If* G *is a linear algebraic group then there is an* F*-torus* $\mathbb{T}$ *inside* G *such that* $\mathbb{T}_{\overline{F}}$ *is a maximal torus.*

Definition 2.5. *A reductive group* $\mathbb{G}$ *over* F *is said to be split or* F*-split if it contains a split maximal torus that is split.*

Example 2.5. $\mathrm{GL}_n, \mathbb{T}_1$ *are split but* $\mathbb{T}_2$ *is not split.*

Proposition 2.1. *For a split reductive group* G*, we can define a root groups* U_α *for* $\alpha \in \Phi(G)$ *over* F*, and similarly other notions such as Borel, parabolic, and so on.*

3 Representation Theory

We will assume that $G = \mathbb{G}(F)$ where $\mathbb{G}$ is a connected reductive group (or for many result G is locally compact Hausdorff topological group so that compact open subgroups form a basis for the neighborhoods of the identity and for any compact open subgroup $K \subseteq G$, G/K is countable).

Definition 3.1. *A smooth representation* (π, V) *of* G *is a complex vector space* V *with a group homomorphism* $\pi : G \to \mathrm{Aut}(V)$ *such that for any* $v \in V$, $\mathrm{Stab}_G(v)$ *is open.*

Proposition 3.1. *The following statements are equivalent for a representation* (π, V) *of* G.

1. *It is smooth.*
2. $V = \bigcup_{\substack{K \subseteq G \\ \text{compact open} \\ \text{subgroup}}} V^K$.
3. *The map* $G \times V \ni (g, v) \mapsto \pi(g)(v)$ *where* V *is endowed with the discrete topology is continuous.*

Definition 3.2. *A morphism of two smooth representation is a* $\mathbb{C}$*-linear map of their vector spaces which is* G*-equivariant.*

Remark 3.1. *The category* $\mathrm{Rep}(G)$ *of smooth representations is abelian together with the morphism in Definition 3.2.*

Definition 3.3. *A smooth representation is irreducible if it contains precisely two subrepresentations.*

Lemma 3.1. *Let* (π, V) *be a smooth irreducible representation of* G*, then* $\dim_{\mathbb{C}} V$ *is countable. If* G *is compact, then* $\dim_{\mathbb{C}} V$ *is finite.*

Proof. Let $0 \neq v \in V$ and $K \subseteq G$ compact such that $v \in V^K$. Since V is irreudcible, $V = \mathrm{Span}_{\mathbb{C}}\{\pi(g)(v) \mid g \in G\}$. However, as K is fixing v, we can also write,

$$V = \mathrm{Span}_{\mathbb{C}}\{\pi(g)(v) \mid g \in G/K\}.$$

As G/K is countable, hence the first assertion. Furthermore, if G is compact, then the latter also follows. □

Theorem 3.1 (Schur's Lemma)**.** *If* (π, V) *is irreducible then* $\mathrm{End}_G(V) \cong \mathbb{C}$.

Proof. Since for $f \in \mathrm{End}(V)$, $\ker f, \mathrm{im}\, f$ are both subrepresentations, hence either f is 0 or bijection. Hence $\mathrm{End}_G(V)$ is a division algebra over $\mathbb{C}$. Note that $f \in \mathrm{End}_G(V)$ is determiend by an image of $v \in V \setminus \{0\}$, hence $\mathrm{End}_G(V)$ is of countable dimension.

Suppose $f \neq c \cdot \mathrm{id}_V$ for some $c \in \mathbb{C}$, then $\{(f - c\,\mathrm{id}_V)^{-1} \mid c \in \mathbb{C}\}$ is linearly independent and uncountable hence a contradiction. □

Definition 3.4. *A smooth representation* (π, V) *is admissible if for any compact open subgroup* $K \subseteq G$*,* V^K *is of finite dimension.*

Proposition 3.2. *A smooth ireducible representation of* G *is admissible.*

Definition 3.5. *Let* H *be a closed subgroup of* G *then the induction of a smooth* H*-representation* (σ, W) *is a* G*-representation* $(\mathrm{Ind}_H^G(\sigma), \mathrm{Ind}_H^G(W))$ *such that*

$$\mathrm{Ind}_H^G(W) := \left\{ f : G \to W \;\middle|\; \begin{array}{c} \forall g \in G, h \in H, f(hg) = \sigma(h) f(g) \\ \exists K(f) \subseteq G \text{ a compact open subgroup} \\ \text{depending on } f, \forall k \in K(f), f(gk) = f(g) \end{array} \right\}.$$

And the action $\mathrm{Ind}_H^G(\sigma)$ *is defined as follows.*

$$\forall g \in G, f \in \mathrm{Ind}_H^G(W), \mathrm{Ind}_H^G(\sigma)(g)(f) = f(\cdot g).$$

Lemma 3.2 (Frobenius Reciprocity)**.** *For* $(\sigma, W) \in \mathrm{Rep}(H), (\pi, V) \in \mathrm{Rep}(G)$*, we have*

$$\mathrm{Hom}_G(V, \mathrm{Ind}_H^G(W)) \cong \mathrm{Hom}_H(\mathrm{Res}_H^G(V), W).$$

That is $\mathrm{Res}_H^G \dashv \mathrm{Ind}_H^G$.

Proof. Exercise to fill the details. The idea is

$$\mathrm{Hom}_G(V, \mathrm{Ind}_H^G(W)) \underset{\alpha_\beta := [V \ni v \mapsto [G \ni g \mapsto \beta(\pi(g))(v)]] \mapsto \beta}{\overset{\alpha \mapsto \beta_\alpha := [V \ni v \mapsto \alpha(v)(1_H)]}{\rightleftarrows}} \mathrm{Hom}_H(\mathrm{Res}_H^G(V), W)$$

□

Definition 3.6. *For H-representation (σ, W), where $H \subseteq G$ is a closed subgroup. The compact-induction is a subrepresentation $(c\text{-}\mathrm{ind}_H^G(\sigma), c\text{-}\mathrm{ind}_H^G(W))$ of $(\mathrm{Ind}_H^G(\sigma), \mathrm{Ind}_H^G)$ such that for any $f \in c - \mathrm{ind}_H^G(W)$, the support of f has a compact image in $H\backslash G$.*

Lemma 3.3. *Both Ind_H^G and $c - \mathrm{ind}_H^G$ are both additive and exact.*

Lemma 3.4. *Let $H, G, (\sigma, W), (\pi, V)$ as in Lemma 3.2. Furthermore, assume that H is open. THen*

$$\mathrm{Hom}_G(c - \mathrm{ind}_H^G(W), V) \cong \mathrm{Hom}_H(W, \mathrm{Res}_H^G(V)).$$

Proof. Exercise. □

Definition 3.7 (Parabolic Induction)**.** *Let $\mathbb{P} = \mathbb{M} \ltimes \mathbb{N} \subseteq \mathbb{G}$ be a parabolic subgroup with Levi decomposition (i.e., $\mathbb{P}, \mathbb{M}, \mathbb{N}$ are subgroups of $\mathbb{G}$ over F such that $\mathbb{P}_{\overline{F}} = \mathbb{M}_{\overline{F}} \ltimes \mathbb{N}_{\overline{F}} \subseteq \mathbb{G}_{\overline{F}}$).*

Let (σ, V_σ) to be a smooth representation of $M = \mathbb{M}(F)$. Then $\mathrm{Ind}_P^G(\sigma) := \mathrm{Ind}_{M \times N}^G \sigma \ltimes \mathrm{triv}$.

Remark 3.2. *Some authors normalize induction by replacing $\sigma(m)$ by $\sigma(m)\delta_p^{2^{-1}}(m)$ where*

$$\delta_p(m) = |\det \mathrm{Ad}_{\mathrm{Lie}(N)}(m)|.$$

Definition 3.8. *An irreducible smooth representation (π, V) of G is called supercuspidal if for any parabolic subgroup $\mathbb{P} = \mathbb{M}\mathbb{N} \subsetneq \mathbb{G}$ and every irreducible smooth representation (σ, V_σ) of M, we have $(\pi, V) \not\hookrightarrow (\mathrm{Ind}_P^G \sigma, \mathrm{Ind}_P^G(V_\sigma))$*

Proposition 3.3. *Let (π, V) be an irreducible smooth representation of G, then there is a parabolic subgroup $\mathbb{P} = \mathbb{M}\mathbb{N} \subseteq \mathbb{G}$ such that a supercuspidal representation (σ, V_σ) of M whose parabolic induction contains π that is*

$$(\pi, V) \hookrightarrow (\mathrm{Ind}_P^G \sigma, \mathrm{Ind}_P^G V_\sigma).$$

4 Moy-Prasad Filtrations

Suppose $\mathbb{G}$ is a split reductive group.

Definition 4.1. *Let $\alpha, \beta \in \Phi(G)$, $X_\beta \in \mathfrak{g}_\beta(F)\backslash\{0\}$ and s_β be the corresponding reflection in $W_G(T)$. w_β is the explicit lift of s_β in $N(T)$ constructed from X_β.*

Remark 4.1. *For the notion of explicit lift see the literature.*

Example 4.1. *For $G = \mathrm{SL}_2$, $X_\beta = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $w_\beta = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$*

Definition 4.2. *A Chevalley system $\{X_\alpha\}_{\alpha\in\Phi(G)}$ is such that $X_\alpha \in \mathfrak{g}_\alpha(F)$ a basis element for each $\alpha \in \Phi(G)$ such that*

$$\mathrm{Ad}(w_\beta)(X_\alpha) = \pm X_{s_\beta(\alpha)}.$$

(The sign can be chosen explicitly)

Example 4.2. *For GL_n and $X_{t_i t_j^{-1}}$*

Definition 4.3 (BT-Triple)**.** *A BT-triplet $(\mathbb{T}, \{X_\alpha\}, X_{\mathrm{BT}})$ consists of*

1. *a split maximal torus $\mathbb{T} \subseteq \mathbb{G}$,*
2. *a Chevalley system $\{X_\alpha\}_{\alpha\in\Phi(G)}$,*
3. $X_{\mathrm{BT}} \in \underbrace{X_*(T)}_{:=\mathrm{Hom}(\mathbb{G}_m,\mathbb{T})} \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$.

5 Moy-Prasad Filtration

Definition 5.1. *For $T = \mathbb{T}(F)$,*

$$T_0 = \{t \in T \mid \forall \chi \in X^*(\mathbb{T}), \mathrm{val}(\chi(t)) = 0\},$$
$$T_r = \{t \in T_0 \mid \forall \chi \in X^*(\mathbb{T}), \mathrm{val}(\chi(t) - 1) \geq r\}.$$

Example 5.1. *For $\mathbb{G} = \mathrm{GL}_n$, we have*

$$T = \{\mathrm{diag}(a_1, \cdots, a_n) \mid a_1, \cdots, a_n \in F^\times\},$$
$$T_0 = \{\mathrm{diag}(a_1, \cdots, a_n) \mid a_1, \cdots, a_n \in \mathcal{O}_F^\times\},$$
$$T_r = \{\mathrm{diag}(a_1, \cdots, a_n) \mid a_1, \cdots, a_n \in 1 + \varpi^{\lceil r \rceil}\mathcal{O}_F^\times\}.$$

Remark 5.1. *For $F = \mathbb{Q}_p$ and $\mathcal{O}_F = \mathbb{Z}_p$, we have the uniformizer $\varpi = p$. In terms of congruence groups of modular forms,*

$$T_k \equiv \{I_n\} \mod p^k\mathbb{Z}_p.$$

Definition 5.2. *Recall that $\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m) \cong \mathbb{Z}$. Hence for $\alpha \in X^*(\mathbb{T})$ and $x_{\mathrm{BT}} \in X_* \otimes_{\mathbb{Z}} \mathbb{R}$, $\alpha(x_{\mathrm{BT}}) \in \mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m) \otimes_{\mathbb{Z}} \mathbb{R}$, hence it corresponds to some r such that*

$$\alpha(x_{\mathrm{BT}})(y) = y^r$$

for some r. Hence, we define $\alpha(x_{\mathrm{BT}}) = r$.

Definition 5.3. *For the root group $U_\alpha = \mathbb{U}_\alpha(F)$ of $\alpha \in \Phi(\mathbb{G}, \mathbb{T})$, we have*

$$x_\alpha : \mathbb{G}_a \xrightarrow{\simeq} \mathbb{U}_\alpha,$$

where $\mathbb{G}_a$ is the additive group such that $\mathbb{G}_a(F) = (F, +)$. Then

$$\mathrm{Lie}(x_\alpha) : \mathbb{G}_a \ni 1 \mapsto X_\alpha \in \mathfrak{g}_\alpha.$$

We then define,

$$U_{\alpha,x,r} := x_\alpha(\varpi^N(\mathcal{O}_F)),$$

where N is the smallest integer such that $\alpha(X_{\mathrm{BT}}) + N \geq r$.

Example 5.2. *For $\mathbb{G} = \mathrm{SL}_2$ and we choose BT-triplet to be*

$$x_1 := \left(\{\mathrm{diag}(t, t^{-1}) \mid t \in F^\times\}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, x_{\mathrm{BT}} = 0 \right).$$

Note that we have

$$\mathbb{Z} \underset{\mathbb{Z}}{\otimes} \mathbb{R} \ni n \mapsto (F^\times \ni t \mapsto \mathrm{diag}(t^n, t^{-n})) \in X_*(\mathbb{T}) \underset{\mathbb{Z}}{\otimes} \mathbb{R}.$$

Let us choose $\alpha \in \Phi(\mathrm{SL}_2) = \{\pm\alpha\}$ where

$$\alpha(\mathrm{diag}(t, t^{-1})) \mapsto t^2, -\alpha(\mathrm{diag}(t, t^{-1})) \mapsto t^{-2}.$$

Then

$$\mathfrak{g}_\alpha = \left\{ \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} \right\}.$$

Hence

$$F \ni y \overset{x_\alpha}{\mapsto} U_\alpha = \left\{ \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} \right\}$$

and

$$\mathbb{G}_a \ni 1 \overset{\mathrm{Lie}(x_\alpha)}{\mapsto} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{g}_\alpha.$$

With these identifications,

$$U_{\alpha,x,r} = x_\alpha(\varpi^{\lceil r - \alpha(x_{\mathrm{BT}}) \rceil}\mathcal{O}_F) = x_\alpha(\varpi^{\lceil r \rceil}\mathcal{O}) = \left\{ \begin{pmatrix} 1 & \varpi^{\lceil r \rceil}\mathcal{O}_F \\ 0 & 1 \end{pmatrix} \right\},$$

$$U_{-\alpha,x,r} = x_{-\alpha}(\varpi^{\lceil r + \alpha(x_{\mathrm{BT}}) \rceil}\mathcal{O}_F) = x_{-\alpha}(\varpi^{\lceil r \rceil}\mathcal{O}) = \left\{ \begin{pmatrix} 1 & 0 \\ \varpi^{\lceil r \rceil}\mathcal{O}_F & 1 \end{pmatrix} \right\}$$

Example 5.3. *We continue the basic setting in Example 5.2. This time we choose* BT-*triplet to be*

$$x_2 := \left(\{\mathrm{diag}(t, t^{-1}) \mid t \in F^\times\}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, x_{\mathrm{BT}} = \frac{1}{4} \right).$$

Observe that for the same α in Example 5.2,

$$t \mapsto \mathrm{diag}(t, t^{-1}) \overset{\alpha}{\mapsto} t^2.$$

Hence we obtain

$$\alpha(x_{\mathrm{BT}}) = \frac{1}{2}.$$

In this case,

$$U_{\alpha,x,r} = x_\alpha\left(\varpi^{\lceil r-\alpha(x_{\mathrm{BT}})\rceil}\mathcal{O}_F\right) = x_\alpha\left(\varpi^{\left\lceil r-\frac{1}{2}\right\rceil}\mathcal{O}\right) = \left\{\begin{pmatrix}1 & \varpi^{\left\lceil r-\frac{1}{2}\right\rceil}\mathcal{O}_F\\ 0 & 1\end{pmatrix}\right\},$$

$$U_{-\alpha,x,r} = x_{-\alpha}\left(\varpi^{\lceil r+\alpha(x_{\mathrm{BT}})\rceil}\mathcal{O}_F\right) = x_{-\alpha}\left(\varpi^{\left\lceil r+\frac{1}{2}\right\rceil}\mathcal{O}\right) = \left\{\begin{pmatrix}1 & 0\\ \varpi^{\left\lceil r+\frac{1}{2}\right\rceil}\mathcal{O}_F & 1\end{pmatrix}\right\}.$$

Hence in this case,

$$U_{\alpha,x_2,0} = \left\{\begin{pmatrix}1 & \mathcal{O}_F\\ 0 & 1\end{pmatrix}\right\}, U_{-\alpha,x_2,0} = \left\{\begin{pmatrix}1 & 0\\ \varpi\mathcal{O}_F & 1\end{pmatrix}\right\}$$

Definition 5.4 (Moy-Prasad Filtration)**.** *Let $r \geq 0$ and x be a* BT*-triplet. Then*

$$G(F) \supset G(F)_{x,r} = G_{x,r} := \langle T_r, U_{\alpha,x,r} \mid \alpha \in \Phi(G)\rangle.$$

Example 5.4. *Let $\mathbb{G} = \mathrm{SL}_2$ and x_1, x_2 from Example 5.2,*

$$G_{x_1,0} = \mathrm{SL}_2(\mathcal{O}_F), G_{x_1,r} = \left\{\begin{pmatrix}1+\varpi^{\lceil r\rceil}\mathcal{O}_F & \varpi^{\lceil r\rceil}\mathcal{O}_F\\ \varpi^{\lceil r\rceil}\mathcal{O}_F & 1+\varpi^{\lceil r\rceil}\mathcal{O}_F.\end{pmatrix}\right\}_{\det=1}.$$

Also from Example 5.3, we see

$$G_{x_2,0} = \left\{\begin{pmatrix}\mathcal{O}_F & \mathcal{O}_F\\ \varpi\mathcal{O}_F & \mathcal{O}_F\end{pmatrix}\right\}_{\det=1}, G_{x_2,r} = \left\{\begin{pmatrix}1+\varpi^{\lceil r\rceil}\mathcal{O}_F & \varpi^{\left\lceil r-\frac{1}{2}\right\rceil}\mathcal{O}_F\\ \varpi^{\left\lceil r+\frac{1}{2}\right\rceil}\mathcal{O}_F & 1+\varpi^{\lceil r\rceil}\mathcal{O}_F.\end{pmatrix}\right\}_{\det=1}.$$

Definition 5.5. *Groups of the form $G_{x,0}$ are called parahoric subgroups of G and $G_{x,r+} = \bigcup_{s>r} G_{x,s}$.*

Proposition 5.1.

1. *$G_{x,r} \lhd G_{x,0}$.*
2. *$G_{x,0}/G_{x,0+}$ is k_F poiints of the reductive group where k_F is the residue field of F.*
3. *For $r > 0$, $G_{x,r}/G_{x,r+}$ is a k_F-vectorspace.*

Example 5.5. *For $\mathbb{G} = \mathrm{SL}_2$ and in Example 5.2,*

$$G_{x_1,0}/G_{x_1,0+} \cong \mathrm{SL}_2(k_F).$$

For $\mathbb{G} = \mathrm{SL}_2$ and in Example 5.3,

$$\begin{aligned} G_{x_2,0}/G_{x_2,0+} &\cong \left\{\begin{pmatrix}\mathcal{O}_F & \mathcal{O}_F\\ \varpi\mathcal{O}_F & \mathcal{O}_F\end{pmatrix}\right\}_{\det=1} \Big/ \left\{\begin{pmatrix}1+\varpi\mathcal{O}_F & \mathcal{O}_F\\ \varpi\mathcal{O}_F & 1+\varpi\mathcal{O}_F\end{pmatrix}\right\}_{\det=1}\\ &\cong \{\mathrm{diag}(t,t^{-1}) \mid t \in k_F^\times\} \cong \mathbb{G}_m(k_F).\end{aligned}$$

Example 5.6. *For $\mathbb{G} = \mathrm{SL}_2$ and in Example 5.2,*

$$G_{x_1,n}/G_{x_1,n+} \cong \mathrm{Mat}_{n\times n}(k_F)_{\mathrm{tr}=0}.$$

For $\mathbb{G} = \mathrm{SL}_2$ and in Example 5.3,

$$\begin{aligned} G_{x_2,\frac{1}{2}}/G_{x_2,\frac{1}{2}+} &\cong \left\{ \begin{pmatrix} 1+\varpi\mathcal{O}_F & \mathcal{O}_F \\ \varpi\mathcal{O}_F & 1+\varpi\mathcal{O}_F \end{pmatrix} \right\}_{\det=1} \Big/ \left\{ \begin{pmatrix} 1+\varpi\mathcal{O}_F & \varpi\mathcal{O}_F \\ \varpi^2\mathcal{O}_F & 1+\varpi\mathcal{O}_F \end{pmatrix} \right\}_{\det=1} \\ &\cong \left\{ \begin{pmatrix} 0 & k_F \\ k_F & 0 \end{pmatrix} \right\} \cong k_F^2. \end{aligned}$$

Furthermore,

$$G_{x_2,1}/G_{x_2,1+} \cong \left\{ \begin{pmatrix} k_F & \\ & k_F \end{pmatrix} \right\}_{\mathrm{tr}=0} \cong k_F.$$

Remark 5.2. *$G_{x,0}/G_{x,0+} \circlearrowright G_{x,r}/G_{x,r+}$ for $r > 0$ by conjugation. This turns out to be linear algebraic action.*

Remark 5.3. *$r_1, r_2 \in \mathbb{R}_{\geq 0}$, we have*

$$[G_{x,r_1}, G_{x,r_2}] \subseteq G_{x,r_1+r_2}.$$

Definition 5.6 (Filtration of Lie algebras)**.** *Let $r \in \mathbb{R}$, we define,*

$$\mathfrak{t}(F) := \mathrm{Lie}(\mathbb{T})(F), \mathfrak{t}_r := \{X \in \mathfrak{Z}(F) \mid \forall \chi \in X^*(\mathbb{T}), \mathrm{val}(d\chi(X)) \geq r\}$$

where $d\chi = \mathrm{Lie}(\chi)$ is the derivative of χ.

We further define,

$$\mathfrak{g}_{\alpha,x,r} := \varpi^{\lceil r-\alpha(x_{\mathrm{BT}})\rceil}\mathcal{O}_F \cdot X_\alpha \subseteq \mathfrak{g}_\alpha(F),$$

and

$$\mathfrak{g}(F)_{x,r} := \mathfrak{g}_{x,r} = \mathfrak{Z}_r \bigoplus_{\alpha\in\Phi(\mathbb{G},\mathbb{T})} \mathfrak{g}_{\alpha,x,r}.$$

Theorem 5.1 (Moy-Prasad Isomorphism)**.** *For $r_1, r_2 \in \mathbb{R}_{>0}$ for $r_1 < r_2 \leq 2r_1$, we have*

$$G_{x,r_1}/G_{x,r_2} \simeq \mathfrak{g}_{x,r_1}/\mathfrak{g}_{x,r_2}.$$

Sketch of the Proof. We have

$$[G_{x,r_1}, G_{x,r_1}] \subseteq G_{x,r_1+r_2} \subseteq G_{x,r_2},$$

hence the quotient should be abelian. □

6 Bruhat-Tits Buildings

Definition 6.1 (non-standard)**.** *The (reduced) Bruaht-Tits building $\mathcal{B}(\mathbb{G}, F)$ of $\mathbb{G}$ over F is such that*

$$\{\text{BT-}\textit{triplets}\}/\sim$$

where the equivalence relation $\sim$ is given by that for two triplets x_1, x_2,

$$\forall r \in \mathbb{R}_{\geq 0}, G_{x_1 r} = G_{x_2, r}.$$

Notation 6.1. *Under the setting of Definition 6.1, it makese sense to write $G_{x,r}$ for each $x \in \mathcal{B}(\mathbb{G}, F)$.*

Remark 6.1. *$\mathcal{B}(\mathbb{G}, F)$ admits an action of G such that for $g \in G$, $G_{gx,r} := gG_{x,r}g^{-1}$ for all $x \in \mathcal{B}(\mathbb{G}, F)$ and $r \geq 0$. That is there is such $y \in \mathcal{B}(\mathbb{G}, F)$ such that $G_{y,r} = gG_{x,r}g^{-1}$ for any g, r as in the above.*

Definition 6.2. *Let (π, V) be an irreducible smooth representation of G. The depth of (π, V) is the smallest non-negative real number r such that $V^{G_{x,r+}} \neq \{0\}$ for some $x \in \mathcal{B}(\mathbb{G}, F)$.*

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